

Unit 8: Training an Artificial Neural Network

What?

This unit focused on the process of training an artificial neural network using **gradient descent**, whose primary goal is to minimize the cost (loss) function while reaching an optimal solution in the fewest possible iterations.

Key concepts covered included:

- **Local and global minima/maxima**, and how optimization algorithms search for the lowest point in the cost landscape.
- **Hyperparameters**, particularly:
 - Learning rate
 - Maximum number of iterations
- The role of the **cost function** in measuring prediction error.

The unit also explored different approaches to gradient descent:

- **Batch (Vanilla) Gradient Descent**
- **Stochastic Gradient Descent (SGD)**
- **Mini-batch Gradient Descent**

For each approach, we analysed the trade-offs between **computational speed, stability**, and **convergence behaviour**.

In the e-portfolio activity, I experimented with different combinations of learning rates and iteration values to observe how these hyperparameters influence model convergence and training performance.

So What?

The practical experiment helped reinforce the theoretical concepts discussed in class. In particular, it was revealing to observe that a **high learning rate combined with a low number of iterations can prevent the algorithm from reaching the global minimum**. Instead of steadily approaching the optimal solution, the model may take excessively large steps and overshoot the minimum.

The results clearly demonstrated that lower learning rates produced a smooth and stable decrease in the cost function, allowing the regression model to gradually converge towards the expected line. In contrast, larger learning rates caused the parameter updates to oscillate or diverge entirely, resulting

in rapidly increasing cost values. This explains why some experimental runs produced extremely large final errors rather than converging to an optimal solution.

Seeing this behaviour firsthand helped me better understand the importance of carefully tuning hyperparameters. While a larger learning rate can speed up training, it may also sacrifice stability and prevent successful convergence.

Now What?

This unit changed my perspective on neural network training by showing that achieving good performance is not simply a matter of increasing computational power or choosing a more complex model. Instead, successful machine learning depends on understanding the interaction between the algorithm, the data, and the hyperparameters used during training.

The gradient descent experiment demonstrated that model performance can be highly sensitive to parameter choices such as learning rate and iteration count. Before this activity, I viewed hyperparameters as technical settings that could be adjusted after selecting a model. However, I now understand that they play a critical role in determining whether a model converges to a useful solution or fails entirely. This insight has encouraged me to adopt a more systematic and evidence-based approach to model development, where experimentation and evaluation are essential parts of the process.

The seminar discussion comparing conventional machine learning models and deep learning techniques further reinforced the importance of selecting methods according to the problem context. The study by Yu et al. showed that CNNs can outperform traditional machine learning models for forecasting distribution costs in supply chain management because they are able to capture complex relationships within the data. However, it also became clear that no single model is universally optimal. A model that performs exceptionally well in one domain may perform poorly in another due to differences in data characteristics, sample size, feature structure, or computational constraints.

In future projects, I will be more critical when choosing machine learning approaches rather than assuming that deep learning models will always provide superior results. I intend to evaluate multiple candidate models, compare their performance using appropriate metrics, and justify model selection based on both predictive accuracy and practical considerations such as interpretability, training cost, and scalability. Additionally, I would like to further develop my skills in hyperparameter optimisation techniques and neural network architecture design to improve my ability to build robust and efficient AI solutions. Overall, this unit has strengthened my understanding that effective machine learning requires not only technical implementation but also informed decision-making throughout the model development process.